

SAM TSANG

Medical Robotics & AI Engineer — MSc, University College London

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PROFILE

Engineer who builds real medical devices and proves they work with measurements — an MR-safe pneumatic actuator for robot-assisted endovascular surgery, an EMG-driven soft exoskeleton, and a vision-controlled prosthetic hand. Strong across mechatronic prototyping (pneumatics, Arduino firmware, drive electronics, 3D printing), robotics software (Python, C++, MATLAB, ROS 2, SLAM) and large-scale statistical analysis in R.

EDUCATION

MSc Medical Robotics & Artificial Intelligence — University College London — Predicted Merit Sep 2025 – Sep 2026

Robotic Systems Engineering · Medical Robotics & Devices · AI for Surgery · Surgical Data Science · ML in Medical Imaging · Robot Vision & Navigation

BEng Biomedical Engineering — University of Reading — Upper Second Class Honours (2:1) Sep 2022 – Jun 2025

EXPERIENCE

Research Student — MR-Safe Surgical Robotics UCL Medical Physics & Biomedical Engineering Oct 2025 – Jul 2026

- **The problem.** MRI is the only modality that images soft tissue in real time without ionising radiation, which makes it attractive for guiding a catheter through a patient's vasculature. Nothing ferromagnetic can go near the scanner bore, so conventional electric motors are ruled out. The project drove the catheter using compressed air alone, keeping every electrically active component in the control room and the in-bore actuator entirely non-metallic and 3D-printed (MR-Safe under ASTM F2503).
- **Built the drive system end to end.** A three-valve solenoid manifold, MOSFET switching electronics on a 24 V rail kept electrically separate from the 5 V logic with flyback protection across each coil, and Arduino firmware running a non-blocking state machine that fires the valves in a phased sequence, converting reciprocating chamber motion into continuous shaft rotation without needing a suction stroke.
- **Designed and ran the test programme** on a purpose-built bench of compressor, regulator, digital pressure gauge and force gauge: torque, shaft speed, thrust and conversion efficiency measured across the full working pressure range with repeats at every point, plus sustained continuous-running trials.
- **Diagnosed the limiting fault by measuring rather than guessing.** Output came in below prediction. Instead of swapping components I instrumented the pneumatic circuit and measured pressure at each stage under load, finding the chamber never saw the regulated supply and that almost the entire loss sat across a single solenoid valve. That inverted the redesign priority from the mechanism to the valve and transmission.

Summer Research Intern — Biomedical Data Science CUHK-Shenzhen, Futian Biomedical Innovation R&D Jul – Aug 2025

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- Analysed **812,883 health assessment records** in R: cleaned and standardised demographic, temporal and clinical variables, then modelled outcomes against season, month, gender and age group using binomial and multinomial logistic regression, reporting odds ratios with 95% confidence intervals.
- Strengthened inference by repeating the modelling across 10 random seeds, averaging estimates and combining p-values via Fisher's method, parallelised with *future.apply*; extended to longitudinal analysis of repeat visits, follow-up intervals and state transitions with questionnaire-level regression and Wilcoxon tests.
- Built a Python semantic screening tool (all-MiniLM-L6-v2 sentence embeddings, cosine similarity) ranking 121 biomedical start-ups against 14 strategic criteria and shortlisting 9; delivered forest plots, trend charts and Excel reporting in ggplot2.

SELECTED PROJECTS

Surgical Workflow Recognition — Cholec80 (103 GB). PyTorch ResNet-18 phase recognition with leakage-safe video-level splits; diagnosed majority-class mode collapse, applied weighted loss, built a timed-fusion model (training loss 1.77 → 0.06), validated with paired Wilcoxon tests. PyTorch · CNNs · Python

Visual SLAM — implementation & ablation. EKF and factor-graph SLAM in MATLAB; systematic ORB-SLAM2 (C++) ablations on KITTI-07 and TUM RGB-D, patching Tracking.cc; Sim(3)-aligned ATE of 2.84 m outdoor / 5.1 cm indoor. C++ · SLAM · MATLAB

Manipulator kinematics, dynamics & planning. Checkpoint-following pipeline for a 5-DOF arm in Python/ROS 2 — rosbag2 extraction, KDL forward kinematics, SLERP interpolation, damped-least-squares IK — plus 7-DOF dynamics derived from first principles. ROS 2 · Python · Kinematics

Webcam-controlled robotic prosthetic hand. Real-time gesture replication on a 3D-printed Brunel Hand 2.0 with five Dynamixel AX-12A motors; OpenCV + MediaPipe tracking mapped to motor commands. *Department showcase selection.* OpenCV · MediaPipe · Dynamixel

EMG-controlled soft exoskeleton. Wearable forearm assistance using McKibben pneumatic muscles triggered by three MyoWare EMG sensors via Arduino and a MOSFET-switched pump; four actuator generations validated for force, range of motion and wearability. EMG · Pneumatics · Arduino

Spooky Eyes — real-time gaze following (BEng dissertation). Caffe SSD face tracking streamed over TCP into a rigged Blender model at ~20 FPS and ~50 ms latency from a commodity webcam; EMA smoothing with deadzone and ease-to-neutral for lifelike motion. OpenCV · Python · Blender

TECHNICAL SKILLS

Programming	Python · C++ · MATLAB · R · SQL
Python libraries	NumPy · pandas · SciPy · Matplotlib · PyTorch · torchvision · OpenCV · MediaPipe · scikit-learn · SentenceTransformers · Pillow · pytest
R libraries	dplyr · tidyr · ggplot2 · ggpubr · rstatix · nnet · future.apply · writexl · patchwork · parallel
Robotics & Control	ROS 2 · SLAM (ORB-SLAM2, EKF, factor graphs) · forward & inverse kinematics · trajectory planning · pneumatics · Arduino / embedded · MOSFET drive electronics · EMG interfacing · real-time control
Hardware & Tools	3D printing · Fusion 360 · circuit prototyping · soldering · Dynamixel servos · Raspberry Pi · multimeter & bench instrumentation · Git · Linux · LaTeX

ADDITIONAL

Languages: English, Cantonese, Mandarin — all native · **Awards:** Silver Medal, UK Mathematics Olympiad · Finalist, University of Reading Project Showcase · **Certificates:** Introduction to Robotics (Coursera) · Python for Data Science (edX)